BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 079 · Project Review and Assessment Feedback

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the purpose of testing a program against expected outputs?

Q2. What can unsafe SQL input let an attacker change?
A. The CPU's clock speed
B. An optical disc's capacity
C. A database query
D. A cable's length

Q3. Can an array be passed to or returned from a subprogram?

Q4. A school updates its anti-malware. “Anti-malware can detect every future threat.” Correct this. Why are other controls needed?

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I Do · Read new knowledge

Key vocabulary

Test evidence — A table should record test input, expected result, actual result and outcome.

Fault diagnosis — Compare expectation and observed behaviour to identify a specific error.

Refinement — Change the faulty instruction and run the affected test again.

Feedback — Use genuine tester or teacher comments to choose improvements, not guessed marks.

Test case — An input and expected outcome used to check correctness.

Bug — A defect in the program.

New learning

Test evidence: A table should record test input, expected result, actual result and outcome. Consider this in a project correction.

Fault diagnosis: Compare expectation and observed behaviour to identify a specific error. Consider this in a quiz total review.

Refinement: Change the faulty instruction and run the affected test again. Consider this in a ticket app test failure.

Feedback: Use genuine tester or teacher comments to choose improvements, not guessed marks. Consider this in a club database feedback.

Worked example

A project correction needs test evidence and fault diagnosis. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].

3. Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].

4. Give a specific point and a linked reason for each feature.

We Do · Practise with guidance

Guided 1

In a quiz total review, explain the purpose of test evidence. [2 marks]

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Worked method: Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].

Guided 2

A ticket app test failure needs fault diagnosis. Explain its role. [2 marks]

Worked method: Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].

Guided 3

A club database feedback uses refinement and feedback. Explain both. [4 marks]

Worked method: Refinement [1]; Change the faulty instruction and run the affected test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the calculation so it prints the price of three £6 items.

```
unitPrice = 6
quantity = 3
total = unitPrice + quantity
print(total)
```

iterate

Question 1

State TWO concepts relevant to this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain how fault diagnosis and refinement help in a café receipt bug. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

Explain the roles of test evidence, refinement and feedback in a pupil scoring app. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A table should record test input, expected result, actual result and outcome.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. To compare its actual behaviour with the requirements and identify faults.

Q2. C — A database query

Q3. Yes. A subprogram can receive an array; a function can return an array.

Q4. Anti-malware detects, quarantines or removes malware. Updates help it recognise threats. It cannot stop every attack, so other controls are needed.

I Do / We Do pages · Guided practice

Q1. Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].

Q2. Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].

Q3. Refinement [1]; Change the faulty instruction and run the affected test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1].

You Do pages · Code First

```
unitPrice = 6
quantity = 3
total = unitPrice * quantity
print(total)
```

Expected result: 18

You Do pages · Independent questions

Q1. Test evidence [1]; Refinement [1].

Q2. Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1]. Refinement [1]; Change the faulty instruction and run the affected test again. [1].

Q3. Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1]. Refinement [1]; Change the faulty instruction and run the affected

test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: